

of BMC operation and related milk storage data are also available over the Internet through mobile phone or PC.

Stellapps' ConTrak solution comprises:

1. Automated temperature measurement using the latest temperature sensor
2. Automated operation of compressor with configurable cut-off and cut-in temperatures
3. Automated agitator operation for efficient cooling and maintaining the milk quality
4. Digital display of temperature, compressor and agitator
5. Manual over-ride for compressor and agitator operations

SmartAMCU milk procurement application

SmartAMCU is an IoT hand-held computing device-based automatic milk collection unit (AMCU). It allows operators to easily manage the milk procurement process, without manual intervention for money management and transactions, thus allowing transparent and easy transactions. The Cloud part of solution facilitates data analysis and storage, control and management by centralised rate-chart push, centralised configuration management, dynamic rate-chart management, analytics of various collection centres based on geographic hierarchy, and even farmer-level views for the entire milk collection activities.

SmartAMCU comprises milk analyser with stirrer, data processing unit (DPU), printer, digital indicator and UPS/inverter. The DPU captures data such as member ID, milk type, fat, solids-not-fat, milk quantity, corrected lactometer reading, temperature, added water, time when quality reading is taken, time when quantity reading is taken, auto/manual mode, and rinsing and cleaning of milk analyser.

Overcoming challenges

"We bootstrapped this initiative with the founders' corpus contribution, when we started in April 2011.

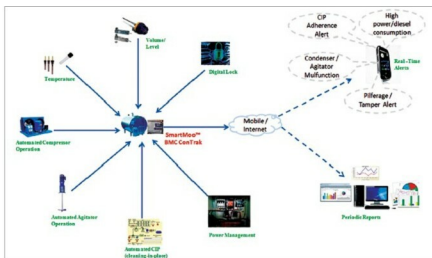


Fig. 2: SmartMoo BMC ConTrak specifications

Subsequently, in 2012, we got angel funded/incubated by IIT-Madras Rural Technology Business Incubator (RTBI)—now merged with IIT-M Incubation Cell. During the 2011-13 period, we also got investments from our friends and family (including our ex-Wipro colleagues and IIT-M alumni)," shared Mukundan.

"We were predominantly in the R&D phase for the first four years, given the intense engineering required to realise the Cloud- and electronics-side products. We started selling seriously into the market about 18 months ago. In this segment, customers expected Cloud software services to be free—a challenge which we are trying to surmount," Mukundan added.

"For some of the parts that we import, overseas vendors had no idea that we were deploying these solutions to the extreme rural areas and that the products had to be rugged. It took a couple of visits from our end to their location and they also visited India to understand the scenario. Moreover, we now have an onsite QC to address hardware related problems," explained Prasanna.

Revenue model

For its consumers, Stellapps has deployed a Capex to Opex model, wherein the customer pays an installation amount in the beginning and then

a monthly instalment depending on the amount of milk being produced. On an average, it charges ₹ 20-40 million from large customers, ₹ 2-10 million from mid-sized dairies and less than ₹ 2 million from small dairies.

The organisation's 70 per cent revenue comes from the milk procurement solution deployed by big centres. The cold chain system, which increases milk's shelf life, brings about 20 per cent, while another 10 per cent is contributed by products used in farms.

The road ahead

Sharing the company's aggressive growth plan, Mukundan said, "Given the unique nature of our solution, the market demand exceeds our execution/production capacity. Right now, we are concentrating onto dairy domain but plan to aggressively expand our business to other emerging markets."

"Through OEM partnership with larger companies, we plan to ride on top of their marketing strength to grow fast outside India. We are also considering setting up channel partners and dealer networks to sell retail-oriented products like animal wearables (activity meters). Onboarding agriculture and dairy veterans who have vast exposure across various departments is our key hiring strategy to get more business," he added. **EFY**